Q4 Reflection and Next Steps: Mira, Nexora Pvt. Ltd. capstone

Mira passed all 12 baseline tests in each of Runs 2 to 5 with no hallucinations, at about $0.0003 per request. This reflection covers what the traces showed, what I would improve, and how I would roll it out.

Trace findings

- **Cost and speed.** In Langfuse, the orchestrator call costs about $0.00008 and takes about 2 seconds. The specialist call costs more ($0.0002 to $0.0003) and takes 1.6 to 3.7 seconds. The whole request takes 10 to 12 seconds from start to finish, so a large share of the wait is outside the model calls, most likely loading the files from Drive and sending the trace. I have not measured that split yet.
- **The evaluator is lenient.** The `risk_relevance` evaluator gave 5.00 to every risk trace it scored. My offline calibration shows why this cannot be trusted alone: the judge agrees with a human on pass/fail only 58% to 68% of the time (kappa 0.23 to 0.27), and it misses wrong task links.
- **Failures the checks missed.** Two intermittent wording errors passed the code checks: a wrong start date in the plan (KL-1) and "no blocked items" in the Sprint 2 update (KL-2). They show that scoring by structure cannot judge prose. The approval step is what stops them reaching a client.

Improvement plan

1. **Code grounding check.** Check every task ID (T001 to T025) and risk ID (R01 to R10) in an answer against the board and register, before the judge runs. This fixes the false fails of judge v2 (KL-9) and catches invented IDs. I have a tested node for it, not yet deployed.
2. **Prompt fixes for KL-1 and KL-2.** Tell the planner that the project start date comes from the timeline file only, and scope status claims to the sprint requested. Add code checks for dates.
3. **Judge v3.** Give the judge the full task board so it can verify reasons, and retest on fresh human-rated items, because the current test set has been used to tune v2.

4. **Second-model comparison.** Run it once the OpenAI allow-list permits a second model (KL-7).
5. **Speed.** Cache the four input files so each request does not download them again.
6. **Attach approval decisions to Langfuse traces.** After the PM approves or rejects in Gmail, send the decision, the reason and the time to decision to the same trace as a score. This would give a first-pass approval rate, show rejections next to the answers they rejected, and let me compare the judge's score with the PM's real decision.

Connection to evaluation and fine-tuning

The evaluation set is the starting point for fine-tuning. The 33 human-rated answers, with the corrected reasons I wrote for the failures, are the first training examples for the Risk Assessor, and the 12 baseline tests plus a fresh, unseen set stay as the exam. Rating those 33 answers took real effort and still gave only fair agreement, so the lesson is that labelling is the main cost. I would fine-tune only after collecting a few hundred corrected examples from real approvals, using the rejection reasons and edited text that PMs already enter in the approval form, and I would use Langfuse annotation queues to collect them.

Privacy considerations

- Project files can hold client data, and every request sends them to OpenAI. Use only the approved provider, on terms that exclude training on our data, and send only the fields a request needs.
- Langfuse traces store full prompts and answers. Restrict access to the project, set a retention period, and keep client names out of test data.
- Nothing leaves without a PM approving it in Gmail. Credentials stay in n8n, and workflow exports are checked for keys, tokens and webhook URLs before they are shared.

Rollout for 10 PMs/TPMs

- **Weeks 1 to 2: two pilot PMs, shadow mode.** They produce plans, risk assessments and reports by hand as usual and compare them with Mira's. Fix the issues found.
- **Weeks 3 to 4: five PMs, approval required.** Mira drafts and the PM approves or rejects with a reason. Add a one-hour training session and a standard input template.
- **Week 5 onward: all ten.** Review the rejection reasons weekly.

- **Go or no-go gates:** at least 80% of drafts accepted without a rewrite, no wrong IDs reaching a client, and a measured time saving on plans and weekly reports.
- **Cost is not the constraint:** ten PMs making 20 requests a week would cost well under $1 a month at about $0.0003 per request. The cost that matters is PM review time.